

CYBERSECURITY

LAB 6

I. Introduction

The hash function is an important element of the message authentication technique. It gets various size input and produces a fixed-size hash value. The hash function uses a compression function repetitively to generate n -bit output. In the digital signature procedure, the hash value uses private and public keys for processing. Message authentication deals with the protection of messages with integrity. It checks the identity of the message sender and non-repudiation of the origin. It checks whether the received messages are originated from the original sender. It ensures that the content of the message is not modified or altered. It also verifies the sequence and timing of the messages. A digital signature is an authentication technique that is used to check the repudiation from the sender's side or the receiver side. The authentication of the digital signature is done on two levels. The sender sends a signed message to the receiver. The receiver compares the computed hash codes with the hash code he got. If both hashes match, he/she can view the message. The most popular hash function is SHA (Secure Hash Algorithm). SHAs are cryptographic algorithms that provide data integrity and - authentication. TLS, SSL, SSH, and PGP applications use SHA. Unlike stream ciphers, which create a long output from a short one, hash functions take a long input and produce a short output, called a hash value or digest.

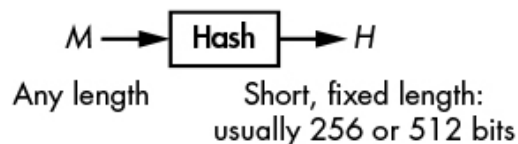


Figure 1 A hash function's input and output

Whitefield Diffie and Martin Hellman were the first to publish a public-key algorithm that solved the problem of key agreement or key exchange. The algorithm is generally referred to as Diffie-Hellman key exchange. The main purpose of the algorithm is to enable two users to securely exchange a key which may then be used for subsequent encryption of messages. The algorithm is limited only to the exchange of keys and is not used for encryption or decryption of messages.

The Diffie-Hellman Key Exchange is a cryptographic protocol exchanging cryptographic keys via a public network. It provides high security of key exchange between entities in an insecure channel. The exchanged keys are used later for symmetric communication using a popular symmetric cipher such as AES, Blowfish, ChaCha20, etc.

DHKE is also called a key agreement protocol. The Diffie-Hellman Key Exchange protocol can be implemented using discrete logarithms (classic DHKE algorithm) or elliptical-curve cryptography (ECDH algorithm). Below is a visualization of key exchange between users.

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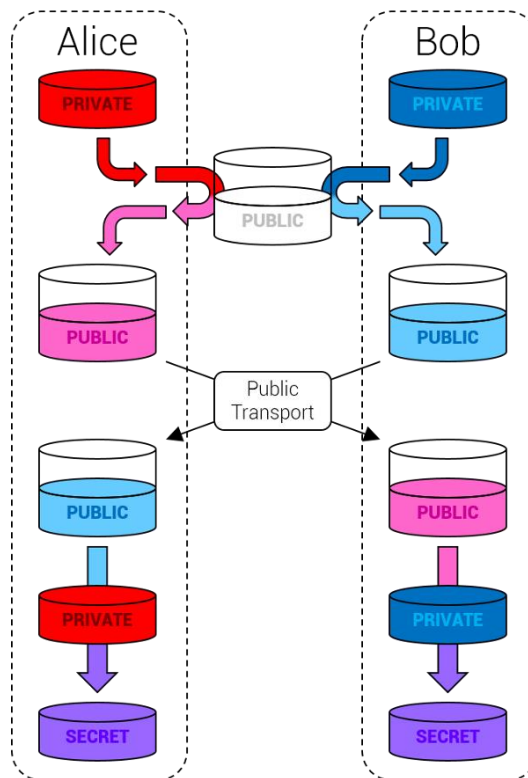


Figure 2 Diffie-Hellman key exchange protocol - source <http://crypto.mdc.io/2012/10/13/public-key-cryptography/>

II. Prerequisites

In this laboratory, the digest functions, the introduction to digital signatures, and key exchange algorithms are analyzed. During the execution of tasks, tools such as Cryptool and OpenSSL library are used. Get acquainted with the following elements:

- In Cryptool look at:
 - Digital signatures/PKI -> Signature Demonstration,
 - Individ. Procedures -> Hash -> Hash Demonstration,
 - Individ. Procedures -> Protocols -> Diffie-Hellman Demonstration,
- Diffie-Hellman key exchange,
- Hash function,

III. Environments

Virtual machines to use in this lab:

- Ubuntu Server,
- Kali,

IV. Tasks

1. Hash function

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Tasks:

- 1.1. Perform the attack on the hash function based on the default files. (Analysis -> Hash -> Attack on the hash value of the digital signature)
- 1.2. Perform the attack again, but change the hash function and the significant bit length (at least two others).

Questions:

- 1.3. How does changing the key length affect the time of the attack?
- 1.4. Does the selection of the hash function affect the time of the collision search task (attack)?
- 1.5. What is the advantage of an attack (finding a collision) whose aim is to modify two documents than one?
- 1.6. Can the hash function be regarded as safe? If so, for which parameters?
- 1.7. What types of problems have been identified for popular hash functions (MD5, SHA)?

2. Key exchange - Diffie-Hellman protocol

Get acquainted with Diffie-Hellman Demonstration available in Cryptool. For demonstration purposes, use values generated in Cryptool. Then the OpenSSL library is used for the practical application of the DH protocol. Use two virtual machines as clients (Kali Linux, Ubuntu) to exchange the keys.

Short description of the next steps:

- Generate public DH key for both clients,
- Based on the public DH key, the clients generate public and private keys,
- The clients must exchange their public keys,
- Based on the public key and private key, the shared secret key is derived.

Tasks:

- a. Generate the Diffie-Hellman common public key for both clients:

```
openssl genpkey -genparam -algorithm DH -out dh_com_pub.pem
```

- b. To display the common key, use:

- a. Encoded form:

```
cat dh_com_pub.pem
```

- b. Text form:

```
openssl pkeyparam -in dh_com_pub.pem -text
```

- c. Based on common DH key, each client should generate their private keys:

```
openssl genpkey -paramfile dh_com_pub.pem -out
```

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dh_client1_key.pem

List the key:

```
openssl pkey -in dh_client1_key.pem -text -noout
```

d. Extract the public key based on the private key:

```
openssl pkey -in dh_client1_key.pem -pubout -out  
dh_client1_pub_key.pem
```

e. Exchange the public keys of the client between them and generate the shared secret key:

```
OpenSSL pkeyutl -derive -inkey dh_client1_key.pem -peerkey  
dh_client2_pub_key.pem -out secret_key1.bin
```

f. Compare the shared secret keys using `cmp` command.

g. Create a hex dump of the shared secrets using `xxd` command.

Questions:

- 2.1. Which key exchange algorithm guarantees higher security? RSA versus DH. Why?
- 2.2. What are the risks for entities using DH key exchange protocol?
- 2.3. What are the other methods to determine a common cryptographic key?
- 2.4. Which element of the DH protocol is not transferred between clients? How does this affect security issues?
- 2.5. Check whether the last point has managed to establish the same key for both clients. Include the content of the agreed secret key in the report.